

- 1) **Step 1:** Search for Windows Powershell & click on “Run As Administrator”
Step 2:

```
PS C:\WINDOWS\system32> wsl --install
A distribution with the supplied name already exists. Use --name to chose a different name.
Error code: Wsl/InstallDistro/ERROR_ALREADY_EXISTS
```

Step 3: Restart Your Computer

After installation, restart your PC when prompted.

Step 4: Complete Ubuntu Setup

After restarting:

1. Ubuntu opens automatically (or search for **Ubuntu** in the Start menu).
2. Wait for the installation to finish.
3. Create a Linux username.

Example:

Enter new UNIX username: nithya

Step 5: Create a Password

You'll then be prompted to create a password.

New password: *****

Retype new password:*****

```
nithya@WIN-482M0QMA856:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
87 packages can be upgraded. Run 'apt list --upgradable' to see them.
nithya@WIN-482M0QMA856:~$
```

Step 7: Install GCC (C Compiler)

```
nithya@WIN-482M0QMA856:~$ sudo apt install build-essential
Installing:
  build-essential

Installing dependencies:
  bzip2      g++-15      libalgorithm-diff-xs-perl  libfakeroot      make
  dpkg-dev  g++-15-x86-64-linux-gnu  libalgorithm-merge-perl  libfile-fcntllock-perl
  fakeroot  g++-x86-64-linux-gnu    libcrypt-dev             libstdc++-15-dev
  g++       libalgorithm-diff-perl  libdpkg-perl             lto-disabled-list

Suggested packages:
  bzip2-doc      debian-tag2upload-keyring  g++-15-multilib  bzip      make-doc
  debian-keyring  g++-multilib              gcc-15-doc       libstdc++-15-doc

Summary:
  Upgrading: 0, Installing: 18, Removing: 0, Not Upgrading: 87
  Download size: 19.3 MB
  Space needed: 73.4 MB / 1024 GB available

Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libstdc++-15-dev amd64 15.2.0-16ubuntu1 [2920 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 g++-15-x86-64-linux-gnu amd64 15.2.0-16ubuntu1 [14.4 MB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 g++-15 amd64 15.2.0-16ubuntu1 [26.7 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 g++-x86-64-linux-gnu amd64 4:15.2.0-5ubuntu1 [968 B]
Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 g++ amd64 4:15.2.0-5ubuntu1 [1100 B]
Get:6 http://archive.ubuntu.com/ubuntu resolute/main amd64 make amd64 4.4.1-3 [197 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdpkg-perl all 1.23.7ubuntu1 [289 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute/main amd64 bzip2 amd64 1.0.8-6build2 [34.3 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute/main amd64 lto-disabled-list all 79 [14.2 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/main amd64 dpkg-dev all 1.23.7ubuntu1 [1062 kB]
```

Step 8: Verify the Installation

```
nithya@WIN-482M0QMA856:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

nithya@WIN-482M0QMA856:~$ |
```

GITHUB REPOSITORY

srinithya2210 / Kernel_Crew_OSSP

+

Code

Issues

Pull requests

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Actions

Projects

Wiki

Security and quality

More

Watch 0

Fork 0

0 stars

0 forks

0 watching

1 branch

0 tags

Activity

Public repository

main

Go to file

+

<> Code

srinithya2210

Initial OSSP folder structure

f7e1b75 · 3 weeks ago

2520030373_Practical	Initial OSSP folder structure	3 weeks ago
2520030373_Skill	Initial OSSP folder structure	3 weeks ago
2520030447_Practical	Initial OSSP folder structure	3 weeks ago
2520030447_Skill	Initial OSSP folder structure	3 weeks ago
2520030483_Practical	Initial OSSP folder structure	3 weeks ago
2520030483_Skill	Initial OSSP folder structure	3 weeks ago
ForgeOS	Initial OSSP folder structure	3 weeks ago

Fork Process

```

#include<unistd.h>
#include<stdio.h>
int main(){
int pid;
pid=fork();
if(pid==0){
printf("Child Process\n");
printf("Child process id is %d\n",getpid());
}
else if(pid>0){
printf("Parent Process\n");
printf("Parent process id is %d\n",getppid());
}
else if(pid<0)
{
printf("Failed process\n");}
}

```

```

nithya@WIN-482M0QMA856:~$ nano skill1.c
nithya@WIN-482M0QMA856:~$ gcc skill1.c
nithya@WIN-482M0QMA856:~$ ./a.out
Parent Process
Parent process id is 364
Child Process
Child process id is 1347

```

exec() process

```

#include<unistd.h>
int main()
{
char *args[]={ "ls", "-l", NULL};
execvp("ls",args);
}

```

```

nithya@WIN-482M0QMA856:~$ nano exec2.c
nithya@WIN-482M0QMA856:~$ gcc exec2.c
nithya@WIN-482M0QMA856:~$ ./a.out
total 220
drwxr-xr-x 3 nithya nithya 4096 Aug 12 15:22 OS_PRACTICE
-rwxr-xr-x 1 nithya nithya 16008 Aug 23 12:57 a.out
-rw-r-xr-x 1 nithya nithya 0 Jul 30 03:49 a1
-rw-r--r-- 1 nithya nithya 83 Aug 12 04:51 a10
-rw-r--r-- 1 nithya nithya 123 Jul 30 03:24 a10.c
-rw-r--r-- 1 nithya nithya 58 Aug 1 05:33 a7
-rw-r--r-- 1 nithya nithya 56 Jul 29 06:01 a8
-rw-r--r-- 1 nithya nithya 39 Jul 29 06:06 a9
-rwxrwxrwx 1 nithya nithya 0 Jul 29 08:23 abc
-rw-r--r-- 1 nithya nithya 30 Aug 1 05:40 b1
-rw-r-xr-x 1 nithya nithya 10 Aug 1 05:58 b2
drwxr-xr-x 2 nithya nithya 4096 Jul 29 13:58 backup
-rw-r--r-- 1 nithya nithya 33 Jul 28 15:10 bio.txt
-rw-r--r-- 1 nithya nithya 0 Jul 29 05:44 data.txt
drwxr-xr-x 2 nithya nithya 4096 Jul 29 05:27 dir1
drwxr-xr-x 2 nithya nithya 4096 Jul 25 05:35 dir2
drwxr-xr-x 2 nithya nithya 4096 Jul 29 05:27 dir3
drwxr-xr-x 2 nithya nithya 4096 Jul 29 05:27 dir4
drwxr-xr-x 2 nithya nithya 4096 Jul 25 05:50 dir5

```

2)

```

#include<stdio.h>
#include<string.h>
int main(){
char input[100];
while(1){
printf("myshell> ");
scanf("%s", input);
if(strcmp(input, "exit")==0){
printf("Goodbye!\n");
break;}
printf("You entered: %s\n",input);
}
}

```

```
nithya@WIN-482M0QMA856:~$ gcc skill2.c
nithya@WIN-482M0QMA856:~$ ./a.out
myshell> Hello
You entered: Hello
myshell> how are you
You entered: how
myshell> You entered: are
myshell> You entered: you
myshell> exit
Goodbye!
nithya@WIN-482M0QMA856:~$ |
```

```
#include <stdio.h>
#include <string.h>
int main()
{
    char buffer[100];
    char ch;
    int i = 0;
    printf("Enter command: ");
    while (1)
    {
        ch = getchar();
        if (ch == '\n')
        {
            buffer[i] = '\0';
            break;
        }
        else if (ch == '\b')
        {
            if (i > 0)
            {
                i--;
                printf("\b \b");
            }
            else
            {
                continue;
            }
        }
        buffer[i] = ch;
        i++;
    }
    printf("%s", buffer);
}
```

```
nithya@WIN-482M0QMA856:~$ nano skill3.c
nithya@WIN-482M0QMA856:~$ gcc skill3.c
nithya@WIN-482M0QMA856:~$ ./a.ouy
-bash: ./a.ouy: No such file or directory
nithya@WIN-482M0QMA856:~$ ./a.out
Enter command: Hello
Hello
Command entered: Hello
Unknown command
```